Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) XOR(1,1) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. 0

B. 1

C. 2

D. Undefined

Correct: A

2. (2 points) Two 100 Ω resistors in series give…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Correct: A

3. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. A live mains circuit

B. Maximum possible voltage

C. No instrument instructions

D. A de-energized suitable circuit

Correct: D

4. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Always adds inputs

B. Stores a clocked bit by definition

C. Measures resistance

D. Selects an input

Correct: D

5. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: Topology follows connections, not visual placement. (4 points).

6. (4 points) Why check rail continuity?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Breadboard rails may be split or connected differently from the assumed layout. (4 points).

7. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Different cases can expose omitted inversions or incorrect branch logic. (4 points).

8. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: The new branch changes equivalent resistance and the voltage distribution. (4 points).

9. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Other paths can alter the reading. (4 points).

10. (4 points) What does a Boolean model omit?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Propagation delay, thresholds, analog transitions, and noise behavior. (4 points).

11. (2 points) A 10 V ideal battery drives a 5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 6 and 8 ohms. Find equivalent resistance.

answer: 14 ohms (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , ohms.

13. (2 points) Two identical 12 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 6 ohms (2 points).

Numerical target 6; absolute tolerance 0.012. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 10 V. Find power.

answer: 20 W (2 points).

Numerical target 20; absolute tolerance 0.04. Accepted units: , W.

15. (2 points) A node has incoming currents 7 A and 2 A and one outgoing current. Find the outgoing current.

answer: 9 A (2 points).

Numerical target 9; absolute tolerance 0.018. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).